

Diffusion-Based Data Augmentation for Skin Lesion Classification

Team members

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Problem and task

Medical image datasets can be heavily imbalanced. Rare skin conditions have few images, so a classifier may learn to ignore those classes.

The project studies whether diffusion-generated images of rare skin lesions can improve classification performance on those classes.

The task is multiclass image classification.

Input: Dermoscopic skin-lesion image resized to 64x64 pixels.

Output: One of the seven HAM10000 lesion classes.

Dataset

HAM10000 contains 10,015 dermoscopic images across seven classes.

The classes and counts are as follows:

	class	images	percentage
0	Melanocytic nevi	6705	66.95
1	Melanoma	1113	11.11
2	Benign keratosis	1099	10.97
3	Basal cell carcinoma	514	5.13
4	Actinic keratoses	327	3.27
5	Vascular lesions	142	1.42
6	Dermatofibroma	115	1.15

Table 1. Number and percentage of images in each HAM10000 skin-lesion class

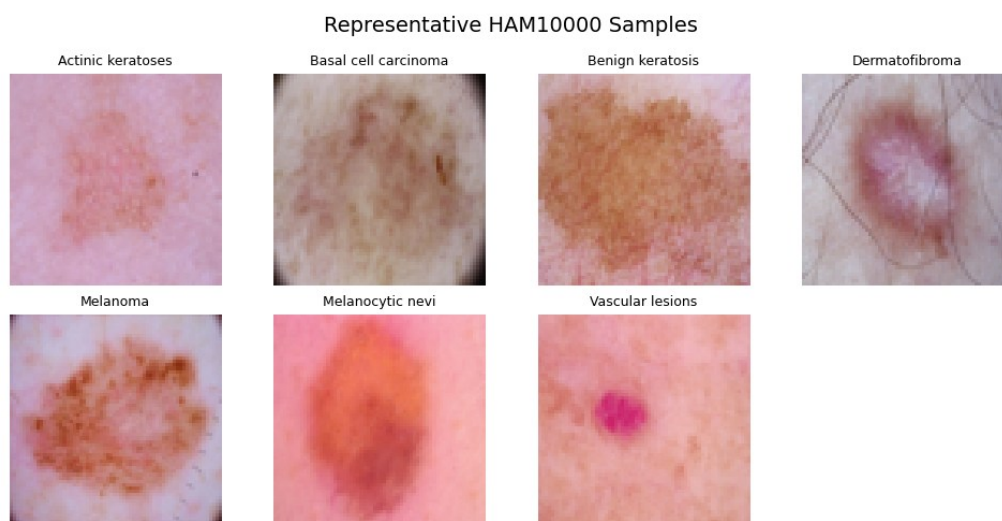


Figure 1. Representative dermoscopic images from the seven HAM10000 skin-lesion classes

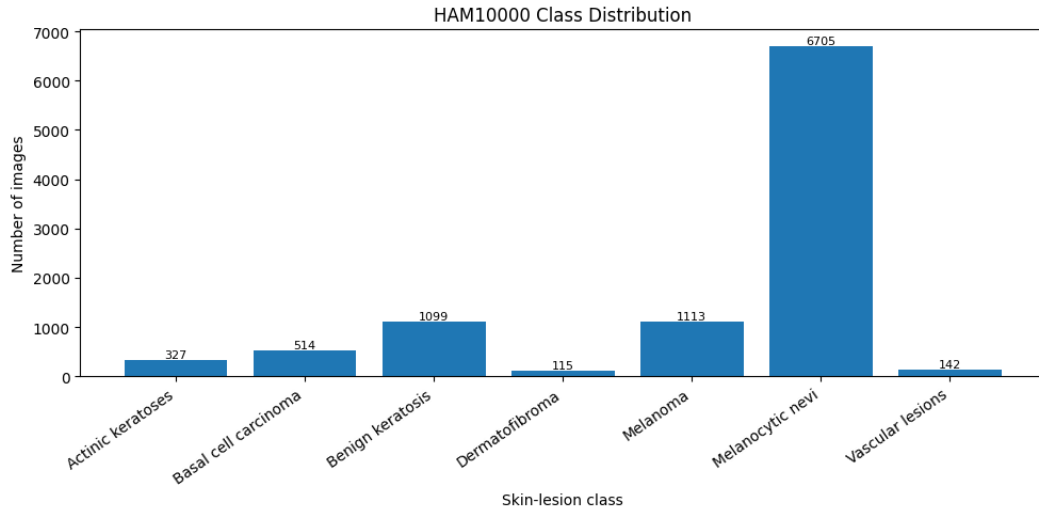


Figure 2. Distribution of images across the seven HAM10000 skin-lesion classes

Dataset access source:

Kaggle:

<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

Original source:

<https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T>

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Planned split

Before model training, the data will be divided into approximately:

- 70% training data
- 15% validation data
- 15% test data

The split will be stratified by class and grouped by lesion ID. A fixed random seed of 42 will be used.

The split contains:

- Training: 6,984 images
- Validation: 1,524 images
- Test: 1,507 images

Class distribution by split:

	split	test	train	validation
class_name				
Actinic keratoses		36	223	68
Basal cell carcinoma		71	338	105
Benign keratosis		185	772	142
Dermatofibroma		22	80	13
Melanocytic nevi		978	4728	999
Melanoma		187	737	189
Vascular lesions		28	106	8

Table 2. Class distribution across the training, validation, and test sets after the stratified group split

Validation metric

Rare-class macro-F1 will be used as the single validation metric.

It is calculated as the average F1-score of the two rarest classes:

$$\text{Rare-class macro-F1} = \frac{F1_{\text{vascular lesions}} + F1_{\text{dermatofibroma}}}{2}$$

This metric is used because the dataset is highly imbalanced. It gives equal importance to vascular lesions and dermatofibroma instead of allowing the majority class to dominate the result.

Baseline model

A ResNet-18 classifier will be trained using only the original real training images. The classifier will predict one of the seven HAM10000 lesion classes.

Diffusion augmentation

A class-conditional diffusion model will generate synthetic images for the two rare classes:

- Vascular lesions
- Dermatofibroma

The synthetic images will be added only to the training set. The validation and test sets will contain real images only.

The experiments will compare:

- Original training set
- Standard augmentation
- Diffusion-based augmentation

Expected results

The baseline classifier may perform poorly on vascular lesions and dermatofibroma because they have few images.

Standard augmentation may provide limited improvement.

Diffusion-based augmentation may improve rare-class macro-F1, but synthetic images may also contain artifacts, so improvement is not guaranteed.

Limitations and ethical considerations

This project is not a clinical diagnostic system.

Synthetic images may contain unrealistic patterns and should not be used for medical decisions.

The HAM10000 dataset may contain demographic and collection biases.

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GitHub repository

<https://github.com/RaneemAlshahrani/Milestone-1-Project-Proposal-and-Experimental-Plan.git>